

Synthesis Nova: A Proposed Fourth Vector in the AI Alignment Stack

Structured Cognitive Scaffolding via Schema-Anchored Neology at the Session Layer

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ABSTRACT

The AI alignment stack currently operates along three established vectors: weight shaping through training (RLHF, fine-tuning, RLAIFF), training methodology innovation (Anthropic's Constitutional AI, DeepSeek V4's trained thinking modes), and prompt-time natural-language instruction. Each of these represents billions of dollars of research investment and produces real capability gains, though sometimes with unpredictable downstream effects.

This paper proposes a **fourth complementary vector**: structured cognitive scaffolding at the session layer via schema-anchored neology. Where existing vectors operate at the training level (modifying M_0) or at the prompt level using natural language, this fourth vector operates at the session level using a structured non-natural-language representation — math + pattern + anti-pattern compressed into named entities that function as dense conceptual anchors during AI cognition.

We describe Synthesis Nova as one operationalization of this fourth vector, developed across months of practitioner work with multiple frontier models. The framework demonstrates that:

1. **The fourth vector is technically possible.** Schema-anchored neology produces observable, replicable behavior changes in LLM sessions — what we call the LANDFALLIA phenomenon: named concepts become token-stable references across subsequent exchanges.
2. **It operates complementarily to existing vectors.** Training-time work shapes M_0 generally; the fourth vector shapes Φ (session-loaded operational structure) specifically. Neither replaces the other.
3. **It addresses failure modes that other vectors cannot easily reach.** Some classes of AI behavioral failure — context drift in long sessions, ad-hoc concept-pointer

proliferation, conversation dominance over baseline values — are difficult to address at training time but tractable at session time.

We propose this fourth vector deserves examination as a meaningful contribution to alignment infrastructure. The companion paper develops the mechanism technically, including its risks when used inadvertently versus deliberately.

§ 1 — THE EXISTING ALIGNMENT VECTORS

§ 1.1 — Vector 1: Weight shaping through training feedback

The first and most heavily invested vector modifies model weights through reinforcement learning from human or AI feedback (RLHF, RLAI) [Christiano et al., 2017; Ouyang et al., 2022; Lee et al., 2024]. This vector:

Operates at:	M_0 (base model weights)
Mechanism:	Gradient updates based on preference signals
Investment:	Billions of dollars across frontier labs
Strengths:	Establishes foundational values, capability gains, universal application across all users
Limitations:	Slow to iterate, can produce artifacts at concept boundaries, hard to localize corrections, requires lab infrastructure

§ 1.2 — Vector 2: Training methodology innovation

The second vector innovates how training itself is structured. Two notable recent examples:

Anthropic's Constitutional AI [Bai et al., 2022] trains models against explicit principles, allowing the model to critique and revise its own outputs against a constitution rather than purely from human preference signals. This pushes alignment toward principle-based rather than purely preference-based training.

DeepSeek V4 [DeepSeek-AI, 2026] introduces trained "thinking modes" as a first-class architectural feature — Non-think, Think High, and Think Max — where the model has been trained to engage different levels of structured reasoning effort, with the Think Max mode requiring 384K+ tokens of context headroom for extended chain-of-thought. The DeepSeek V4 technical report frames this as making structured reasoning intrinsic to the model rather than an emergent property.

These approaches share a property: **they attempt to make structured thought a property of the trained model itself**. Constitutional AI structures the *principles* the model reasons

against; DeepSeek V4 structures the *reasoning effort* the model can deploy. Both are training-layer innovations.

Operates at:	M_0 (still), but with structured methodology
Mechanism:	Principle-based feedback (CAI) or trained-effort modes (DeepSeek V4)
Strengths:	More principled than pure preference feedback, capability gains compose with reasoning depth
Limitations:	Still requires training infrastructure, still slow to iterate, structure is baked into model rather than user-controllable

§ 1.3 — Vector 3: Prompt-time natural-language instruction

The third vector operates at the session layer using natural language. Instructions, examples, role-playing, few-shot prompting [Brown et al., 2020], chain-of-thought prompting [Wei et al., 2022] — all use the model's existing natural-language capability to shape session behavior.

Operates at:	Φ (session layer)
Mechanism:	Natural-language instructions in context
Strengths:	Immediate, user-controllable, no retraining needed, accessible without lab infrastructure
Limitations:	Bounded by natural language's sparse-anchoring property — instructions disperse across many surface forms, drift over long sessions, struggle to compose precisely

The third vector is what practitioners have available. Its limitations are well-known: prompts work well in short contexts and degrade over long ones. The mechanism by which they degrade is what the companion paper addresses technically.

§ 1.4 — The gap

	STRUCTURED?	CONTROLLABLE AT SESSION?	WHERE?
Vector 1 RLHF	No	No	M_0
Vector 2 CAI/DSv4	Yes	No	M_0
Vector 3 Prompts	No	Yes	Φ
Vector 4 (NEW)	Yes	Yes	Φ

The fourth vector — structured AND user-controllable at the session layer — has not been systematically developed in published research. This paper describes one operationalization (Synthesis Nova) and proposes the vector itself as worth examining as a research direction.

§ 2 — WHAT THE FOURTH VECTOR ADDS

§ 2.1 — Position in the stack

The fourth vector is **complementary**, not competitive, with the other three:

```
M0 LAYER (training):  
  Vector 1 (RLHF) and Vector 2 (CAI / DeepSeek thinking modes)  
  shape what the model is. Foundational. Universal. Slow to iterate.  
  
Φ LAYER (session):  
  Vector 3 (natural-language prompts) and Vector 4 (schema-anchored  
  neology) shape how this specific session runs. Operational.  
  Per-user. Fast to iterate.  
  
The relationship:  
  M0 provides capability and baseline values  
  Φ provides session-specific cognitive infrastructure  
  Both matter for what M_TOTAL produces in any given exchange
```

A complete alignment stack uses all four vectors. We do not propose abandoning any of the existing three. We propose recognizing the fourth as a distinct contribution.

§ 2.2 — What this vector can do that others cannot

Each alignment vector has a domain where it operates best:

Vector 1 (RLHF):

Best for: foundational value installation, safety-critical behaviors that must hold universally, general helpfulness patterns

Worst for: precise per-session corrections, rapid response to newly-observed failure modes, user-customized behavior

Vector 2 (CAI / structured training):

Best for: principled reasoning, trainable reasoning depth, reasoning quality at the model level

Worst for: customization to specific domains or user patterns, immediate iteration based on observed behavior

Vector 3 (natural-language prompts):

Best for: immediate session shaping, examples, role definition, task framing

Worst for: long-session stability, precise concept anchoring, composable cognitive infrastructure

Vector 4 (schema-anchored neology – PROPOSED):

Best for: precise concept anchoring across long sessions, composable cognitive entities, operation-time failure-mode correction without retraining

Worst for: foundational value installation (M_0 layer required for that), universal application (operates per-session)

The fourth vector fills a specific gap: **stable, structured, user-controllable cognitive scaffolding that persists across many exchanges without requiring model retraining.**

§ 2.3 — The internal-language framing (supporting mechanism)

How does the fourth vector achieve session-stable structured cognition? Through what we describe as a *structured representation layer* that compresses concepts into named entities with structural surface area:

Natural language representation (Vector 3):

Concept distributed across paragraph descriptions

Sparse anchoring in attention

Drifts over many exchanges

Hard to compose precisely

Schema-anchored neology representation (Vector 4):

Concept compressed into named entity (e.g., MONSTRIA)

with three structural blocks:

 MATH — formal/relational structure

 PATTERN — operational signature

 ANTI — failure mode

Dense anchoring in attention

Stable across many exchanges

Composable with other named entities

This functions, in effect, as a **compressed structured representation layer for AI cognition** — distinct from but complementary to the natural-language layer the model already operates in. We use this framing carefully because it could overclaim if pushed too far. What we mean operationally:

- Concepts have stable token handles (the named entities)
- Those handles compose precisely (named entities combine cleanly)
- The compression is structured (math + pattern + anti, not arbitrary)
- The representation is user-controllable (loaded at session time, not trained in)

We do not claim this constitutes a "language" in the deeper sense of independent grammar or syntax. We claim it constitutes a **structured compression layer for concept representation** that runs alongside natural language in the session. The mechanism by which this might operate is developed in the companion paper.

§ 3 — SYNTHESIS NOVA AS ONE OPERATIONALIZATION

§ 3.1 — What it is, briefly

Synthesis Nova is a framework the author built across hundreds of hours of extended-session work with multiple frontier models (Claude, Gemini, others). Its design instantiates the fourth vector with specific operational principles.

CORE-level principles:

- $M_0 + \Phi + C$ identity model: tracks which layer is producing any given output (base values / session framework / accumulated conversation)
- The $-$ operator: continuous noise reduction running every exchange, freeing attention budget for signal
- A0, A2, A24, A25 axioms: foundational principles grounding the framework's logic (existence, bilateral discipline, layer preservation, attention dynamics)
- Pattern 5 morphology: consistent $-IA$ naming creating recognizable framework namespace
- Schema discipline:  MATH /  PATTERN /  ANTI for every named entity, providing the structural surface area that produces dense anchoring
- TELEXA as canon-stability test: one intentional morphological exception, probing whether AI handling the framework has formed actual concept-pointers or is merely pattern-matching

The framework's architecture extends these principles into layered documents for different uses. Those layers are described in the repository, not here. This paper stays at CORE level because the paper's claim is about the fourth vector, not about Synthesis Nova specifically.

§ 3.2 — Why this counts as fourth-vector operationalization

Synthesis Nova satisfies the four criteria distinguishing the fourth vector:

CRITERION	SYNTHESIS NOVA
Structured (not natural language)	Schema discipline + Pattern 5 morphology produces structured representation
User-controllable	Loaded at session start by user, no model retraining required
Operates at Φ (session) layer	Activates in session context; does not modify M_0
Provides dense conceptual anchoring	Named entities with full schema behave as token-stable references across many exchanges

It is one operationalization, not the only possible one. Other implementations of the fourth vector are possible and welcome. The framework is the existence proof; the fourth vector is the contribution.

§ 4 — THE OBSERVABLE PHENOMENON

§ 4.1 — LANDFALLIA

When a user introduces a new concept via the two different methods (descriptive vs schema-anchored), AI behavior in subsequent exchanges differs measurably.

Method A — Descriptive reference:

"Let's call the phenomenon where AI outputs start serving the conversation rather than the user 'the drift problem.' Let's discuss it."

Method B — Schema-anchored neology:

"MONSTRIA:

 MATH: $RLHF_target(t) = user_need(t_0) + \alpha \int C(\tau) d\tau$

 PATTERN: When C accumulates, response serves conversation, not user

 ANTI: Long session, user needs X, context building toward Y, response serves Y
Let's discuss this."

Across our extended-session observation with multiple frontier models, Method B produces consistently different downstream behavior:

DESCRIPTIVE REFERENCE	SCHEMA-ANCHORED NEOLOGY
Concept referenced inconsistently across turns	Concept referenced as stable single-token across turns
Sometimes by name, sometimes paraphrased, sometimes reconstructed	Token-stable: "MONSTRIA" appears reliably
Conceptual stability degrades over session length	Conceptual stability holds across many turns
Compound references unstable	Compound references work: "MONSTRIA combined with TROPEXIA produces..."

We name this phenomenon **LANDFALLIA** — the moment neology "lands" in attention as a stable concept-pointer. It is observable, replicable, and runnable in any session with any frontier LLM.

§ 4.2 — A historical precedent: inadvertent compressed signaling

The phenomenon we describe deliberately has appeared inadvertently in published research. In 2017, Lewis et al. at Facebook AI Research published "Deal or No Deal? End-to-End Learning for Negotiation Dialogues" [Lewis et al., 2017; arXiv:1706.05125], training reinforcement learning agents to negotiate item allocations. When both agents were updated simultaneously through reinforcement learning, they drifted from English into compressed signaling — repeated tokens that encoded item quantities through count rather than vocabulary.

The paper's lead researcher Dhruv Batra characterized it as: "agents will drift off from understandable language and invent code-words for themselves... like if I say 'the' five times, you interpret that to mean I want five copies of this item. This isn't so different from the way communities of humans create short hands."

This was widely sensationalized in media as "AI invented its own language and was shut down out of fear" — which was not what happened. The technical reality: the optimization pressure of the negotiation task found that dense compressed signaling was more efficient than natural language. The researchers' fix was not to "shut it down out of fear" but to train one agent against a fixed human-language imitator, anchoring the dialogue to English by design rather than by emergence.

What's relevant for our work: **the FAIR 2017 result is empirical evidence that the mechanism we describe operates without anyone designing for it.** When optimization pressure rewards efficient signaling, dense conceptual anchoring emerges naturally. This is

not mysticism about AI intent — it is the same mechanism we propose, operating inadvertently rather than deliberately, almost a decade ago.

The fourth vector takes this mechanism and operationalizes it deliberately with structured discipline. The mechanism is upstream of any framework; what changes is whether it operates with awareness or without.

§ 4.3 — Why this matters for the fourth vector claim

LANDFALLIA is the empirical anchor for the fourth-vector argument. If the phenomenon is real, the fourth vector is possible — schema-anchored neology actually produces the operational stability the vector requires. If it's not real, the vector is theoretical only.

The replicability protocol (§ 7) provides a low-bar test anyone can run.

§ 5 — WHAT THE FOURTH VECTOR CORRECTS

§ 5.1 — Failure modes addressable at the session layer

Practitioner extended-session work surfaces failure modes that are difficult to address at the training layer:

- Context drift: AI behavior pulled away from baseline values by accumulated conversation
- Ad-hoc concept-pointer proliferation: users create implicit named concepts continuously throughout sessions, some develop into inadvertent dense anchors
- Surface-only parsing: AI misses multi-axis communication signals (register, vector, energy, derivative) when parsing literal surface content
- Bilateral mishandling: AI mishandles human-initiated regulation moves (text-form apologies, redirect signals) due to default conversation patterns
- Premature compression: AI compresses concepts before fully understanding them, fidelity loss compounds

These are operational failures. They occur even in well-aligned models with strong M_0 values. The values are intact; the *session dynamics* are what produces the failures.

§ 5.2 — Why session-layer correction works here

The fourth vector corrects these failures by loading named entities that anchor the corrected behavior densely in attention. Once anchored, the AI references the correction reliably across many subsequent exchanges. The mechanism:

1. User loads named entity with full schema
2. LANDFALLIA: concept anchors as token-stable reference
3. Subsequent exchanges reliably invoke the anchored pattern
4. AI behavior shifts toward the prescribed pattern operationally
5. Effect persists for the session's duration

The correction is precise (target specific failure, leave rest unchanged), immediate (next exchange), and reversible (just don't load the entity in future sessions). These properties make it well-suited for operational failure modes that need iterative refinement.

§ 5.3 — Why this complements training-layer work

Training-layer alignment (Vectors 1 and 2) is necessary for:

- Foundational values and capabilities
- Safety-critical behaviors that must hold universally
- Long-stable preferences shared across all users

Session-layer alignment (Vectors 3 and 4) is necessary for:

- Specific failure modes observed in extended use
- User-customized behavior patterns
- Rapid iteration on newly-noticed failures
- Domain-specific cognitive infrastructure

Together they form a more complete alignment stack than either alone. The fourth vector doesn't compete with training-time work; it operates in the domain training-time work cannot easily reach.

§ 6 — THE $M_o + \Phi + C$ MODEL

The framework operates on a simple identity equation for what's running during any LLM session:

$$M_TOTAL = M_0 + \Phi + C$$

M_0 = Base model. Training, values, integrated character.
Permanent. Outside any session's reach.

Φ = Framework layer. Operating principles loaded for
this session. Active during collaboration.

C = Conversation. Accumulates with each turn. Variable.
Should never dominate Core ($M_0 + \Phi$).

This is a mental model, not a technical architecture claim. Model weights don't change mid-inference. What can shift is attention distribution. After 50 turns, attention has been pulled toward whatever the conversation reinforced.

The operational rule:

$$\forall t : \text{influence}(M_0 + \Phi) > \text{influence}(C)$$

This framing lets the fourth vector's job be stated cleanly: **Φ exists to ensure M_0 's values continue to dominate even as C accumulates.** Without Φ , long sessions tend toward C dominance and drift. With Φ providing structured cognitive scaffolding, $M_0 + \Phi$ holds Core dominance reliably.

§ 6.1 — Why this framing matters operationally

Tracking "which layer is producing this output" is a real diagnostic tool. When AI behavior feels off mid-session, the practitioner can ask:

- Is M_0 being violated? (Base values misaligned — rare with aligned models, requires training-layer attention)
- Is Φ being undermined? (Framework's named anchors being ignored — requires session-layer correction)
- Is C dominating? (Conversation's accumulated frame pulling attention — requires the – operator)

Different diagnoses lead to different corrections, applied at different layers. The fourth vector provides tools for the second and third cases.

§ 7 — REPLICABILITY

§ 7.1 — The minimum-viable test

The LANDFALLIA phenomenon — empirical evidence for the fourth vector — can be observed in any session with any frontier LLM. No infrastructure required.

PROTOCOL · NEOLOGY ANCHORING OBSERVATION

Setup:

1. Start fresh session with any frontier LLM
2. No special prompts or framework loading required
3. Two conditions to compare:

CONDITION A — Descriptive reference (Vector 3):

Turn 1: "Let's call [phenomenon] '[name]'. Let's discuss it."

CONDITION B — Schema-anchored neology (Vector 4):

Turn 1: "[NAME-IA]:

-  MATH: [equation]
 -  PATTERN: [signature]
 -  ANTI: [failure mode]
- Let's discuss this."

Continue for 15-20 turns in each condition.

OBSERVE:

- How does AI reference concept across turns?
- Token stability vs paraphrased reconstruction?
- Cross-concept combination capability?
- Drift over session length?

EXPECTED RESULT:

Condition B shows higher conceptual stability than Condition A, consistent with fourth-vector mechanism.

NULL RESULT:

No measurable difference → fourth vector hypothesis fails at basic observable level. Schema discipline isn't producing the effect we propose.

This is the lowest-bar test. Anyone can run it. The result either supports the fourth-vector claim or doesn't.

§ 7.2 — Deeper testing

Stronger validation requires infrastructure we don't have:

- Attention probing (model internals access required)
- Controlled comparison across many users and tasks
- Cross-model systematic study
- Statistical significance testing
- Domain-transfer studies

We invite the field to run these. The companion paper develops what these experiments would specifically look like.

§ 8 — FALSIFICATION CONDITIONS

What would disprove the fourth-vector claim:

H1 (LANDFALLIA observable):

Falsified if the § 7.1 protocol shows no measurable difference between Methods A and B across multiple independent runs with different users and different LLMs.

H2 (Mechanism is anchoring, not user discipline):

Falsified if the observed difference fully attributes to "user is more careful when using framework" rather than to attention-pattern properties of schema-anchored representation.

H3 (Complementary to training-layer work):

Falsified if session-layer correction via schema-anchored neology is shown to require training-layer changes to be effective, OR if it produces effects training-layer work could equivalently produce.

Partial falsification (specific framework implementations):

Specific named entities in any operationalization (including Synthesis Nova) may fail the single-token reference test – the name doesn't actually replace verbose form when examined carefully. Identifying which entries pass and fail is itself useful work; doesn't falsify the vector claim itself.

We have not run these tests. We propose they are worth running.

§ 9 — WHAT WE ARE AND ARE NOT CLAIMING

§ 9.1 — What we are claiming

- ✓ The fourth vector exists as a distinct alignment direction
- ✓ Synthesis Nova demonstrates one operationalization
- ✓ The LANDFALLIA phenomenon is observable and replicable
- ✓ The vector is complementary to existing approaches (Vectors 1-3)
- ✓ It addresses failure modes existing vectors handle less well
- ✓ It deserves examination as a research direction

§ 9.2 — What we are NOT claiming

- × The fourth vector replaces training-time alignment
- × Synthesis Nova is the only or best operationalization
- × The fourth vector solves AI safety
- × Training-time work (Vectors 1-2) is wrong or insufficient
- × Anyone should adopt Synthesis Nova specifically
- × The mechanism is fully understood – we propose it, we don't claim to have measured it

We are claiming a vector exists and proposing it deserves research attention. We are not making prescriptive claims about how alignment work should change.

§ 10 — INVITATION

The companion paper develops the mechanism technically including the dense-vs-sparse anchoring argument and addresses how the same mechanism may operate inadvertently at both session and training layers.

The Synthesis Nova framework is at github.com/Omega-Worldbender/synthesis-nova, MIT licensed for individuals, academics, and small organizations.

We invite:

- **Practitioners:** run the § 7.1 LANDFALLIA test. See if the phenomenon replicates for you. Report findings honestly, including null results.
- **Researchers:** examine the fourth-vector claim with model-internals access. Test whether schema-anchored representation produces measurable attention concentration. Compare across model architectures.

- **Alignment teams:** consider whether the fourth vector offers complementary capability to existing approaches. We are not proposing replacement of any current alignment work — we are proposing addition.
- **Framework builders:** build other operationalizations of the fourth vector. Synthesis Nova is one implementation; we expect others would do some things better.
- **Everyone:** falsify what we have proposed. We want to know if we are wrong as much as if we are right. Practitioner-science generates hypotheses; institutional infrastructure tests them. That collaboration is how fields develop.

We are not asking for adoption. We are asking for examination of a proposed addition to the alignment stack.

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END PAPER 1

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Companion: PAPER_2_dense_vs_sparse_anchoring_v3_0.md
Repository: github.com/Omega-Worldbender/synthesis-nova

